GENUS CANNOT INCREASE ALONG FINITE MORPHISMS THAT ARE ISOMORPHISMS OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label lemma-genus-goes-down names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Higher cohomology of zero-dimensional support sheaves vanishes. Varieties, Lemma \label{varieties-lemma-chi-tensor-finite}. Proof lines 3855-3857: the cokernel F of O_X -> f_*O_Y is supported in finitely many closed points, hence has vanishing higher cohomology.
- Vanishing of H^2 on Noetherian schemes. Cohomology, Proposition \label{cohomology-proposition-vanishing-Noetherian}. Proof lines 3859-3861: H^2(X, G) = 0 for the coherent sheaf G on the Noetherian scheme X.
- Cohomology commutes with affine morphisms. Cohomology of Schemes, Lemma \label{coherent-lemma-relative-affine-cohomology}. Proof lines 3865-3867: H^1(X, f_*O_Y) = H^1(Y, O_Y) because f is finite, hence affine.

Source reference. Original lemma 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Prerequisite pointer invoked at proof line(s) 3874; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Prerequisite pointer invoked at proof line(s) 3870; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 3868; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. Let $f : Y \to X$ be a finite morphism such that there exists a dense open $U \subset X$ over which f is a closed immersion. Then*

$$\dim_k H^1(X, \mathcal{O}_X) \geq \dim_k H^1(Y, \mathcal{O}_Y)$$

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Proof. Consider the exact sequence

$$0 \to \mathcal{G} \to \mathcal{O}_X \to f_*\mathcal{O}_Y \to \mathcal{F} \to 0$$

of coherent sheaves on X. By assumption $\mathcal{F}$ is supported in finitely many closed points and hence has vanishing higher cohomology (Varieties, Lemma 0AYT). On the other hand, we have $H^2(X, \mathcal{G}) = 0$ by Cohomology, Proposition 02UZ. It follows formally that the induced map $H^1(X, \mathcal{O}_X) \to H^1(X, f_*\mathcal{O}_Y)$ is surjective. Since $H^1(X, f_*\mathcal{O}_Y) = H^1(Y, \mathcal{O}_Y)$ (Cohomology of Schemes, Lemma 089W) we conclude the lemma holds. □